

Final Project Detailed Instructions

1 Software Requirement Specification

Format: PDF.

Structure:

1. Introduction
 1. Aim of the document
 2. Overview of the defined system
 3. HW e SW requirements
 4. Related systems (at least 2), Pros and Cons.
2. User Stories (3 per member)
3. Functional Requirements (3 per member)
4. Use Cases:
 1. Overview Diagram (1)
 2. Internal Steps (1 per member)

2 Storyboards

Quantity: min 2 screens per member, covering all the major functionalities described in SRS, developed using Draw.io or similar

3 Design

- Class diagram:
 - 1 VOPC per member.(analysis)
 - 1 design-level diagram per member (e.g. that includes patterns, or specific solutions that improve the engineering level of the system)
- Design patterns: 1 different pattern per member. Possibly try to apply the pattern within the context of the project.
- Activity diagram: 1 per member.
- Sequence diagram: 1 per member.

- State diagram: 1 per member.

4 Testing

- Develop at least 3 test cases per person. In each test (class) file, please report (via Java comments) the name of the person in charge.

5 Code

- ~4K LOC
- Similar functionality implemented with two GUIs.
- Exceptions: at least 2 per member (do not just catch and back-propagate the exceptions, but properly handle them. Possibly define your own error logic by means of exceptions)
- Be able to show that Svn(or Git) + SonarCloud is correctly installed in one of your computer and it is able to analyze your project for rule violations. No rule must be violated (no smells, no vulnerabilities, no bugs). This will be checked during the exam.
- The application has to run in 2 different modalities: demo-version, full-version
 - in the demo-version the application provides all the functionalities you decided to support BUT all the data are only stored IN-MEMORY, NOTHING has to be stored on any persistency layer. In this sense, as all the entities are kept in memory, if the application is turned off all the data are lost
 - in the full-version the application provides all the functionalities you decided to support AND the data can be stored on some persistency layer. Further detail in the next requirement.
- At least one DAO shall be provided in two versions DMBS and file system.

6 Video

- A 1 to 2 minutes recorded video with audio of the developed system performing the expected functionalities. *.mpeg

Notes

1. System.out* CLI-related smells are allowed.
2. Deliverables should be compliant with HMW constraints.
3. Code must be compliant with deliverables. Discrepancies can exist and they should be pointed out in such artifacts.
4. The project is one per team. The grade is shared across team members. All team members must submit the entire project.

5. The project submission procedure will be communicated (only) to the students subscribed to a specific exam date.